

PROJECT 5 — BI DASHBOARD MINI PROJECT

Use Case Proposals

Healthcare SME (51–250 employees) | AI Implementation Roadmap

Prepared for: Chleo, CEO | Date: March 2026 | Confidential

SME Constraints & Selection Rationale

Three use cases were selected for Chleo's organisation. Every candidate was evaluated against four SME-specific constraints before being included:

No dedicated data team → Models must be trainable with standard Python libraries and public datasets	Budget-sensitive → Free-tier APIs, open-source tooling, measurable ROI within 1–3 months	Compliance-critical → PHI exposure minimised; de-identified or synthetic data preferred for PoC	Change-management light → Workflows must integrate into existing staff routines, not replace them
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UC
1

No-Show Prediction

Reduce missed appointments using a transparent, explainable ML model

Feasibility High	ROI Timeline 4–8 weeks	PHI Exposure None	Complexity Low–Med	Explainability High	Priority 1st
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What it does

A gradient-boosted classification model (XGBoost or LightGBM) is trained on historical appointment records. Each morning it generates a ranked list of today's and tomorrow's appointments by predicted no-show probability. The scheduling team uses this list to prioritise reminder calls, fill cancelled slots, or apply double-booking logic to high-risk slots.

Why this use case for a healthcare SME

- No-shows cost an average primary care practice 14–18% of available appointment capacity. For an SME, that is directly recoverable revenue.
- The model requires zero patient PHI — it uses only scheduling metadata: age group, appointment lead time, chronic condition flags, SMS reminder status, and historical no-show rate. All can be de-identified before model training.
- XGBoost produces feature importance scores, meaning staff can see exactly why a patient is flagged (e.g. 'long lead time + no SMS confirmation'). This directly answers Chleo's transparency concern.
- No EHR integration required at PoC stage — the model runs on a CSV export from the scheduling system.

Technical implementation

Component	Detail
Dataset	Kaggle: Medical Appointment No-Shows (110,526 rows, 14 features)
Model	XGBoost classifier — scikit-learn pipeline with SMOTE for class imbalance
Features	Age, gender, scholarship, hypertension, diabetes, alcoholism, SMS received, lead time (days), day of week, historical no-show rate
Output	No-show probability score (0–1) + top 3 contributing features per appointment
Serving	Daily CSV report or n8n-triggered dashboard refresh — no real-time API needed at PoC
Monitoring	LangSmith dataset: 200 annotated appointments used to evaluate model calibration weekly

Dashboard tie-in: The BI dashboard will show no-show rate by age group, lead time distribution, and model accuracy over time — all sourced from the Kaggle dataset.

UC
2

Patient Intake Email Triage Automation

Auto-classify and route incoming patient requests with a monitored LLM

Feasibility High	ROI Timeline 2–4 weeks	PHI Exposure Low	Complexity Low	Explainability High	Priority 2nd
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What it does

An n8n workflow monitors the patient inbox (Gmail or Outlook). When a new email arrives, it is passed to an LLM (Claude Haiku or GPT-4o-mini) with a structured classification prompt. The model returns a JSON response: category (appointment request, prescription query, test results, complaint, other), urgency (routine / urgent / emergency), and a suggested response draft. The email is then automatically routed to the correct team queue. Every classification is logged in LangSmith.

Why this use case for a healthcare SME

- Reception and admin staff at SMEs spend 2–4 hours per day manually triaging inbound patient emails. This use case reclaims that time without requiring any clinical decision-making by the AI.
- The LLM never makes a clinical recommendation — it categorises and routes only. This keeps the system well outside regulated clinical AI territory and avoids HIPAA complexity at PoC stage.
- Using Claude Haiku or GPT-4o-mini keeps per-email cost under \$0.001 — negligible at SME volumes (typically 50–200 emails/day).
- The n8n workflow is fully auditable: every email, classification, and routing decision is logged with timestamp, model response, and confidence level.

- LangSmith monitoring allows the practice manager to review a weekly accuracy report — showing how many classifications were correct, overridden by staff, or flagged as uncertain.

Technical implementation

Component	Detail
Trigger	n8n Gmail / Outlook node — fires on new email in inbox
LLM	Claude claude-haiku-4-5 (Anthropic) or GPT-4o-mini — structured JSON output via function calling
Prompt	System prompt defines 5 categories + urgency levels; includes few-shot examples; output schema enforced
Output	JSON: {category, urgency, confidence, draft_reply} — routed to department Slack channel or ticketing system
Fallback	Confidence < 0.75 → flagged for human review with 'uncertain' tag; never auto-routed
Dataset	Synthetic intake emails generated with Faker (200 records, 5 categories) — used for LangSmith eval
Monitoring	LangSmith traces every call; weekly eval run scores precision/recall per category

n8n PoC tie-in: This is the core workflow demonstrated in the proof-of-concept — an 8-node n8n flow from email trigger through LLM classification to Slack notification, all visible and auditable.

UC
3

Clinical Note Summarisation (Monitored)

LLM-assisted SOAP note generation with full LangSmith observability

Feasibility Medium	ROI Timeline 6–10 weeks	PHI Exposure Medium	Complexity Medium	Explainability Medium	Priority 3rd
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What it does

A clinician dictates or pastes free-text consultation notes into a simple web form. The text is sent to an LLM with a structured SOAP (Subjective, Objective, Assessment, Plan) extraction prompt. The model returns a formatted clinical note draft. The clinician reviews, edits, and approves before anything enters the EHR. The LLM never stores the input — it processes and returns immediately. Every interaction is logged in LangSmith for monitoring.

Why this use case for a healthcare SME

- Clinical documentation accounts for 35–55% of a clinician's working day. Even partial automation that reduces note-drafting time by 30% has a direct impact on patient capacity and staff burnout.

- HUMAN IN THE LOOP is hardcoded: the AI produces a draft; the clinician must review and sign off before it goes anywhere. The AI never writes to the EHR directly. This is the architecture that satisfies both regulatory requirements and Chleo's transparency concerns.
- This use case is the most powerful demonstration of LangSmith monitoring — Chleo can see every prompt sent to the AI, every response returned, and every instance where the clinician overrode the AI. That is full observability.
- At PoC stage, synthetic consultation transcripts are used. No real PHI is processed. Rollout to live notes requires a BAA with the LLM vendor, which is documented in the cost estimate.

Technical implementation

Component	Detail
Input	Free-text transcription (dictated or typed) submitted via a minimal web form (Streamlit or HTML)
LLM	Claude Sonnet 4 (Anthropic) — higher reasoning capability justified for clinical context
Prompt	System prompt enforces strict SOAP format; instructs model to flag uncertainty rather than guess; includes refusal behaviour for non-clinical inputs
Output	Structured SOAP note with S / O / A / P sections; uncertain fields flagged with [REVIEW REQUIRED]
Human loop	Clinician must click 'Approve' before note is available for EHR copy-paste; no auto-write
PHI handling	LLM API call: no data retention (Anthropic zero-retention tier for BAA accounts); no logging of raw text to LangSmith — only metadata and output structure
Dataset	50 synthetic consultation transcripts (Faker + domain vocabulary) — LangSmith eval set for SOAP accuracy scoring

LangSmith tie-in: This is the flagship monitoring demonstration. The LangSmith dashboard will show live traces of every summarisation request — inputs, outputs, latency, and the weekly eval score against the synthetic benchmark set.

Use Case Comparison & Selection Rationale

The three use cases were selected to demonstrate the full AI value stack to Chleo: a predictive model (no-show), an automation workflow (intake triage), and a generative AI application (note summarisation). Together they illustrate that AI in healthcare can be measurable, auditable, and incrementally adoptable.

	UC1: No-Show Prediction	UC2: Intake Triage	UC3: Note Summarisation
AI Type	Predictive ML	Generative AI (LLM)	Generative AI (LLM)

PHI in PoC	None	Low (email metadata)	None (synthetic only)
Staff impact	Scheduling team	Reception / admin	Clinicians
Implementation risk	Low	Low	Medium
Monitoring method	Model accuracy metrics	LangSmith classification eval	LangSmith SOAP eval
Dashboard visibility	Primary (full dataset)	Supporting (volume metrics)	Supporting (accuracy gauge)
SME budget fit	Free OSS (scikit-learn)	< €30/month API cost	~€50/month API cost

Dataset Selection & Alignment

All datasets used in this project are publicly available, de-identified, or synthetically generated. No real patient PHI is processed at any stage of the PoC.

Dataset	Source	Use Case	Role in Project
Medical Appointment No-Shows	Kaggle (joniarroba)	UC1	Primary training & evaluation dataset for no-show prediction model. 110,526 rows, 14 features. Powers the BI dashboard.
Synthetic Patient Intake Emails	Generated (Faker + Python)	UC2	200 synthetic emails across 5 categories (appointment, prescription, results, complaint, other). LangSmith eval set for intake classifier.
Synthetic SOAP Transcripts	Generated (Faker + domain vocab)	UC3	50 synthetic consultation transcripts with ground-truth SOAP notes. LangSmith evaluation benchmark for note summarisation accuracy.
WHO Global Health Expenditure	WHO Observatory (public)	Dashboard	Sector-level healthcare spend data for dashboard context visualisation. No PHI. Used in executive summary section.

Dataset generation scripts for the synthetic datasets (UC2, UC3) will be committed to the /data/raw/ directory alongside the Kaggle CSV. Faker seed is fixed for reproducibility.